STAFF REVIEW · H446 249 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Showing low, high, mid, comparisons and any bound updates.

2. O(n log n).

3. n values; storing it needs O(n) space.

4. Left shift 00010110 << 2 = 01011000 (eight-bit, zero-filled). AND 00110110 ∧ 00001111 = 00000110.

I Do — source worked steps

1. Predict first — One scan has eight iterations, giving linear growth O(n).

2. Explain the answer — Nested loops perform 8×8=64 inner iterations, giving quadratic growth $O(n^2)$.

We Do — guided CHECK

1. Predict first
The linear count doubles from 8 to 16.
Explain the answer
Quadratic count rises from 64 to 256, multiplying by four; this difference grows with n.

You Do — source mark schemes

1. [4] O(1) [1], O(n) [1], $O(n^2)$ [1], O(log n) [1].

2. [6] Each of n elements read and written [2], so O(n) time [1]; allocates n extra positions [2], so O(n) auxiliary space [1].

3. [6] Merge sort can scale better to large n [2]; O(n log n) vs $O(n^2)$ [1]; in-place sort can suit small data or tight memory [1]; typically O(1) auxiliary space for in-place quadratic sorting [1]; compare time-space trade-off [1].

Exit Ticket — source answer

1. False. Time and auxiliary-space complexity are different measurements.